



Chitosan-graft-poly(vinyl acetate) for wood-adhesive applications

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ABSTRACT

Poly(vinyl acetate), PVAc, adhesives are commonly used for wood bonding; however, they are fossil-based and the final products usually do not have a sufficient water resistance for more durable applications. In this study we prepared an adhesive formulation by grafting VAc from chitosan using emulsion polymerization, chitosan-graft-PVAc. Thereby, we could decrease the fossil-based content of the adhesive and at the same time significantly improve the water resistance. Chitosan by itself has very good bonding properties as a wood adhesive, especially regarding water resistance; however, very low solid contents of the adhesive formulation can be achieved due to a very high viscosity of chitosan adhesives. In our chitosan-graft-PVAc adhesives, we explored two chitosan samples with different molecular weights, by using as-received chitosan and hydrolyzing it to a lower molecular weight. The chitosan fractions in the adhesives prepared with a higher molecular weight chitosan were 15, 20 and 25 wt%. However, due to the high viscosity, a solid content higher than 17 wt% could not be achieved for these adhesives. Sufficient bond strengths were achieved when the adhesive was applied in 122 g/m² solid spread rate. In order to decrease the viscosity, we used hydrolyzed chitosan, with a lower molecular weight, to allow for a higher adhesive solid content, 34 wt%, and for a higher chitosan fraction, 40 wt%. In the adhesive with 40 wt% chitosan and 17 wt% solid content, all VAc was grafted from chitosan. This decreased the molecular mobility of the chains, leading to a lower susceptibility to plastic creep in the adhesive which contributes to the final bond strength. The dry and wet strengths of the specimens bonded with adhesives containing chitosan were higher than the strength of the specimens bonded with the reference PVAc adhesive.

1. Introduction

One of the commonly used adhesives for wood bonding is poly(vinyl acetate) (PVAc). It is used both in industrial- and home applications. PVAc adhesives have many advantages, e.g., a satisfactory dry strength and room temperature setting. However, products bonded with the most common PVAc adhesives have poor water- and creep resistance, mainly due to the thermoplastic character [1]. The water resistance can be improved by using two-component PVAc adhesives with hardening resins, isocyanates or complex-forming salts [2]. Another drawback of PVAc adhesives is the resource used for their production, which is fossil-based. Research is being conducted aiming to investigate how the creep- and the water resistance of PVAc adhesives can be improved, and/or how to partially replace fossil-based VAc with biobased alternatives [3–8]. Grafting VAc from a biopolymer is a method to

incorporate a biobased component in the PVAc adhesive and to combine the properties of both polymers. Since the polymers are bonded covalently, the grafted copolymers can achieve improved properties compared with physical blends of the two polymers. Grafting vinyl monomers from biobased materials has been extensively investigated, e.g., grafting from cellulose and pulp [9–16], and other biobased polymers such as chitosan and chitin [17–21]. It has previously been reported that the grafting of vinyl monomers from starch can improve the properties of starch-based wood adhesives [22–28]. Since the polymerization of VAc proceeds by a free radical polymerization, the vast majority of reported polymerizations include purging of the reaction system with argon or nitrogen to remove oxygen, since it is known to act as an inhibitor. Common initiators used for grafting from cellulose and similar polymers contain cerium(IV), such as cerium ammonium nitrate (CAN) which was used in this study. Oxygen has been reported as an

Abbreviations: Non-hydrolyzed chitosan, NH-CS; Hydrolyzed chitosan, H-CS; Poly(vinyl acetate), PVAc; Chitosan-graft-poly(vinyl acetate), CS-g-PVAc.

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environment-friendly co-oxidant for CAN [29], which is beneficial in large-scale productions such as the wood-adhesive production. Although grafting VAc from chitosan has been reported previously [30–33], to the best of our knowledge, these polymers have not been evaluated as wood adhesives.

Chitosan is an amino-functional biobased polymer, and it consists of β -(1,4)-linked 2-amino-2-deoxy-D-glucopyranose (glucosamine) and 2-acetamido-2-deoxy-D-glucopyranose (*N*-acetyl glucosamine). Chitosan is obtained by deacetylation of chitin, which is produced by crustaceans and other living organisms. Annually, living organisms produce 10 billion tons of chitin [34]. Chitin is insoluble in common organic and diluted aqueous solvents, and it is deacetylated to produce chitosan, a more soluble derivative. Chitosan is insoluble in most organic solvents and in water at neutral and alkaline pH, but it is soluble under acidic conditions due to the protonation of the amino side-groups. It is used in various fields, e.g., in agriculture, water treatment, and the pharmaceutical- and food industries [34]. As a wood-adhesive component, chitosan has shown promising results [35–41] and it can be envisioned that the use of chitosan may bring a more environmentally-friendly system as the fossil-based content is reduced. In the grafting polymerizations of VAc, chitosan contributes in the reaction as a grafting site, but also as a surfactant [31].

In this study, we have investigated the grafting of VAc from chitosan in order to lower the fossil-based content in a PVAc adhesive and to improve the water resistance of bonded specimens since chitosan adhesives show good water resistance [37,41]. We have hydrolyzed chitosan to reduce its molecular weight and thereby also lowering the viscosity of chitosan solutions [42]. Consequently, the viscosity of the adhesives can be decreased, and therefore the solid content can be increased. All polymerizations were performed in air atmosphere, which is beneficial for large-scale polymerizations.

2. Experimental

2.1. Materials

Low molecular weight chitosan (with a viscosity of 92 mPas and degree of deacetylation of 75–85 %), vinyl acetate (VAc, containing 3–20 ppm hydroquinone as inhibitor, ≥ 99 %), acetic acid (Reagent-Plus®, ≥ 90 %), cerium(IV) ammonium nitrate (CAN, ≥ 99.99 % trace metals basis), azobisisobutyronitrile (AIBN), and sodium hydroxide (NaOH, reagent grade, ≥ 98 %, pellets, anhydrous) were purchased from Sigma Aldrich. Hydrochloric acid (HCl, AnalaR Normapur, 37 %), and ethanol (EtOH, AnalaR Normapur, 96 %) were purchased from VWR.

Cascol® 3304, a poly(vinyl acetate)-based (PVAc) wood adhesive, with 50 wt% solid content, was kindly supplied by AkzoNobel Adhesives AB and used as a reference. Cascol is a thermoplastic wood adhesive suitable for indoor applications. Beech-wood veneers (sliced veneers) were purchased from Holgers Stugmaterial AB, Sweden.

2.2. Methods

2.2.1. Hydrolysis of chitosan

Chitosan (6 g, with 9 % moisture content) was added to 0.5 M HCl (525 mL) in small portions under vigorous stirring with a magnetic stirrer and left to dissolve at 25 °C for 1 h. Afterwards, 74 mL of HCl (37 %) was added (in order to achieve 2 M HCl) and left to react at 60 °C for 1 h. The reaction was quenched by immersing the round-bottom flask in an ice-water bath and subsequent neutralization of the reaction mixture with sodium hydroxide (NaOH) solution (30 wt%). The hydrolyzed chitosan (H-CS) was precipitated with ethanol (equal volumes of the reaction mixture and EtOH), filtered through a 0.45 μ m nylon filter, and then dispersed in deionized water (500–600 mL water). The remaining salt was removed by centrifugation (10 min, 6000 rpm), after which H-CS was freeze-dried.

2.2.2. Emulsion polymerizations

For polymerizations, both non-hydrolyzed (NH-CS) and hydrolyzed chitosan (H-CS) were used, according to the recipes shown in Table S1 in the Supporting information. When NH-CS was used, it was dissolved in 10 wt% acetic acid solution (aq) overnight at 25 °C. When H-CS was used, it was dissolved in 10 wt% acetic acid solution (aq) for 2 h at 25 °C prior to the reaction. Both NH-CS and H-CS were stirred with a magnetic stirrer. After dissolving NH-CS/H-CS, the initiator solution (CAN) was added and left to stir for 15 min. Afterwards, VAc was added, and the reaction mixture was left to stir for another 10 min. The reaction mixture was placed in an oil bath and left to react at 60 °C. After specific times (shown in Table 1), AIBN dissolved in VAc was added, the temperature was increased to 75 °C, and the mixture was left to react for 30 min. This step was conducted as a burnout, to allow any unreacted VAc to polymerize. The names of the prepared adhesives, their target solid contents and chitosan-VAc ratios are shown in Table 1.

2.2.3. Reference adhesives

A commercial PVAc, NH-CS and H-CS were used as reference adhesives. For the preparation of chitosan reference adhesives, NH-CS and H-CS were dissolved overnight in 10 wt% acetic acid solution (aq), with 17 wt% solid content.

2.2.4. Characterization

Fourier Transform Infrared Spectroscopy (FTIR): The measurements were performed on a Spectrum 100 FTIR instrument with a single reflection ATR system, an MIR-TGS detector, an MKII Golden Gate and a diamond crystal (Graseby Specac Ltd, Kent, England). Spectra were recorded as an average of 16 scans at 4.0 cm^{-1} , in the range of 600–4000 cm^{-1} .

Solid content and VAc conversion: To access the solid content, samples of polymerized adhesives were dried at room temperature overnight, and then at 105 ± 5 °C for 1 h, following the European Standard EN 827 [43]. The solid content and the VAc conversion were determined gravimetrically. The VAc conversion was calculated according to equation (1) [31]:

$$\text{Conversion (\%)} = (\text{weight of dried sample} - \text{weight of chitosan}) / \text{weight of added monomer} \quad (\text{equation 1})$$

Grafting efficiency: PVAc homopolymer was extracted in acetone.

Table 1

Names of the samples/adhesives prepared by grafting vinyl acetate (VAc) from chitosan, with target ratios of components, target solid contents, and reaction times.

Sample name ^a	Chitosan used ^b	Chitosan-VAc fraction ^c (wt%)	Target solid content (wt%)	Reaction time (h)
NH-CS 15-17	NH-CS	15–85	17	2
NH-CS 20-17	NH-CS	20–80	17	2
NH-CS 25-17	NH-CS	25–75	17	2
H-CS 25-17	H-CS	25–75	17	4 ^d
H-CS 25-34	H-CS	25–75	34	4 ^d
H-CS 40-17	H-CS	40–60	17	4 ^d

^a Letters in the sample names correspond to the chitosan sample used (non-hydrolyzed chitosan, NH-CS, or hydrolyzed chitosan, H-CS). The first number in sample names represents the chitosan fraction in the total solid amount, and the second number represents the total solid content (SC), in percentages.

^b NH-CS: non-hydrolyzed chitosan; H-CS: hydrolyzed chitosan.

^c Based on the target solid amount.

^d Reactions were also performed for 2 h, but these samples exhibited too low VAc conversions.

Samples of polymerized adhesives (100 mg) were left in acetone (10 mL) overnight in order to dissolve PVAc homopolymer. After centrifugation (15 min, 6000 rpm), soluble and insoluble fractions were separated and dried. The grafting efficiency was calculated as the percentage of grafted PVAc to the total amount of polymerized VAc.

Thermogravimetric analysis (TGA): The measurements were performed on a Mettler Toledo TGA/DSC1 instrument. Samples (10 mg) were heated from 30 to 600 °C at a rate of 10 °C/min, under (1) nitrogen and (2) oxygen flow of 50 mL/min.

Differential Scanning Calorimetry (DSC): The measurements were performed on a Mettler Toledo DSC 5. Samples (8 mg) were analyzed using the following program: heating from 25 to 110 °C, isotherm for 10 min, cooling to -20 °C, isotherm for 10 min, and heating to 300 °C, under nitrogen flow of 50 mL/min. The heating and the cooling rate were 20 °C/min. Analyzed data were obtained from the second heating cycle. Measurements were performed for NH-CS, H-CS, PVAc-g-CS and reference PVAc films dried overnight at 25 °C, acetone-soluble and acetone-insoluble fractions of PVAc-g-CS adhesives which were previously prepared for the analysis of the grafting efficiency.

Viscosity: The viscosity was determined for NH-CS, H-CS and adhesives used for bonding veneers. NH-CS and H-CS (1 wt%) were dissolved in 10 wt% acetic acid solution. The viscosity of the adhesive was determined after polymerization reactions. The measurements were performed on a cone and plate viscometer CP1 (TQC Sheen, The Netherlands) at 20 °C using a 0-10 P cone and 750 rpm rotation speed.

2.2.5. Tensile shear strength

The bond strength of the prepared adhesives was tested according to the modified European Standard EN 205 [44]; wood substrates were thinner than standard requirements. Beech-wood veneers (100 mm × 20 mm × 1.5 mm) were conditioned in standard atmosphere (23 ± 2 °C temperature and 50 ± 5 % relative humidity) before bonding. The absolute moisture content, determined gravimetrically, was 7 % and the average density was 0.73 g/cm³. Adhesives were applied and spread between two veneers with 20 mm overlap (Fig. S1 in the Supporting information). The adhesive amounts applied between veneers were determined based on their solid contents, by aiming for the same solid adhesive amounts in all samples (61 and 122 g/m²). The adhesives with 17 wt% target solid contents were applied in two layers, with a 20 min interval. Applied amounts and number of layers are shown in Table S2 in the Supporting information.

The specimens were pressed at 120 °C and 1.7 MPa for 2.5 min in a Carver hot-press. After pressing, the specimens were conditioned according to the European Standard EN 204 [45] (Table 2).

The tensile shear strength of bonded veneers was determined on a low force universal testing machine (Instron 5566) with a 10 kN load cell and a crosshead speed of 1 mm/min. The tensile shear strength was calculated as the ratio between the maximum force before failure and the bonded area. The results are presented as average values of five samples, with standard deviations presented as error bars. T-test, with α level 0.05, was performed in order to determine statistical differences between data sets; the letters in Figures indicate the significant differences between data sets, and p-values are shown in Tables S3 and S4 in the Supporting information.

Table 2
Conditioning methods for bonded veneers samples.

	Conditioning method	Durability class
Dry strength	7 days in standard atmosphere ^a	Dried
Wet strength	7 days in standard atmosphere ^a , 4 days in water at 20 ± 5 °C	Wet

^a Standard atmosphere: 23 ± 2 °C temperature and 50 ± 5 % relative humidity.

3. Results and discussion

3.1. Hydrolysis of chitosan

Chitosan was hydrolyzed under acidic conditions in order to decrease its molecular weight. With a lower molecular weight, the viscosity of adhesives was lower, and, therefore, the solid content could be increased. The change in chitosan molecular weight after hydrolysis was confirmed by the change in the viscosity of the chitosan solutions [42]. Unfortunately, it was not possible to determine the chitosan molecular weight by size exclusion chromatography at this point since chitosan is not soluble in solvents commonly used for liquid chromatography. Nonetheless, the goal of this work was to analyze the effect of different chitosan molecular weights on the polymerization and adhesive performance, which has been done. The viscosity of the non-hydrolyzed chitosan (NH-CS) solution was 0.24 Pa·s. After hydrolysis, the viscosity of the hydrolyzed chitosan (H-CS) solution was 0.13 Pa·s. The deacetylation of chitin is commonly performed under alkaline conditions, but it could occur also under acidic conditions, depending on the reaction parameters such as acid concentration and temperature [46]. Aljbou et al. hydrolyzed chitosan in HCl (2 M) and reported a decrease in molecular weight together with an increase in the degree of deacetylation [47]; however, the reaction temperature was higher than the temperature used in our study. Funahashi et al. reported that treatment of chitin in HCl (1 M) at 85 °C for 2 h decreased its molecular weight, but not the degree of acetylation [48]. During hydrolysis in this study, the degree of acetylation was (mostly) unaffected. The FTIR spectra of NH-CS and H-CS (Fig. 1) had peaks characteristic for chitosan [49]. The spectra were normalized against the peak at 1421 cm⁻¹, which corresponds to methylene units (-CH₂-) since this peak should not be affected by hydrolysis. The ratio between the primary amine (1585 cm⁻¹) and acetyl groups (1646 cm⁻¹) was almost unaffected, showing that the degree of acetylation in both samples was similar. The broad peak around 3300 cm⁻¹, corresponding to the N-H and O-H stretching, was increased and peaks around 900-1100 cm⁻¹, corresponding to the C-O-C bridges, decreased. This was expected since during hydrolysis the polymer chains are cleaved, which leads to a lower number of C-O-C linkages between glucose units, and increased number of hydroxyl groups at newly formed chain ends.

3.2. Emulsion polymerization

The reaction mixture in an emulsion polymerization is a complex system, and many factors affect the outcome of the polymerization. The results by Fernandez et al., when investigating how the grafting of a mixture of VAc-methyl acrylate from cellulose using emulsion polymerization is affected by the order of addition of reactants, shows that the highest monomer conversion and percentage of grafted polymer were achieved when the initiator was added before the monomers, with 15 min mixing time for the cellulose-initiator mixture [11]. Therefore, in this study, CAN was first added to the chitosan solution and left to mix for 15 min before VAc was added. Solid content, VAc conversion, grafting efficiency and viscosity of the resulting adhesive dispersions are shown in Table 3. Since the polymers were not soluble, it was not possible to determine the VAc conversions by NMR; therefore, they were assessed gravimetrically, based on the determined solid contents (assuming homogeneous samples). When NH-CS was used, all polymerizations had VAc conversion above 90 % with 2 h reaction time, but H-CS decreased the apparent rate of polymerization, and VAc conversions after 2 h were only between 65 and 75 %. This suggests that the polymerization kinetics is affected by the molecular weight of chitosan, possibly altering the stabilization of the emulsion polymerization. With a longer reaction time (4 h), the VAc conversions in all polymerizations were above 90 %. Table 3 shows parameters obtained for adhesives with VAc conversions above 90 % since only those adhesives were used for wood bonding. Adhesives with a lower VAc conversion were not used for

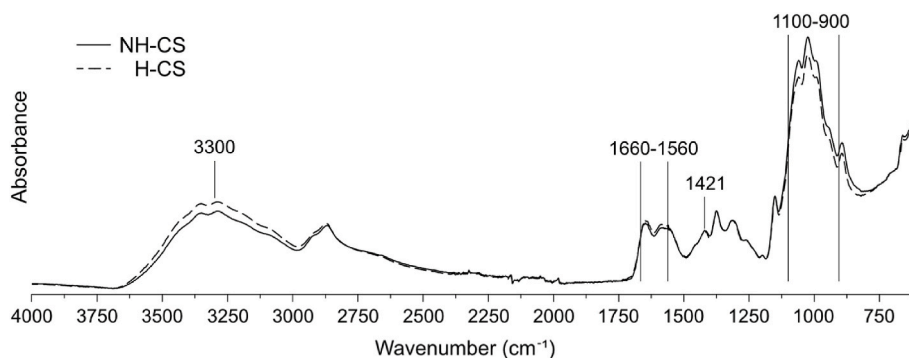


Fig. 1. FTIR spectra of non-hydrolyzed chitosan (NH-CS) and hydrolyzed chitosan (H-CS).

Table 3

Measured solid contents, conversions and grafting efficiencies of vinyl acetate (VAc), and viscosities of prepared chitosan-graft-PVAc adhesives (CS-g-PVAc).

Sample ^a	Measured solid content ^b (wt%)	VAc conversion ^c (%)	Viscosity (Pa·s)	Grafting efficiency (%)
NH-CS_15-17	16	98	0.5	49
NH-CS_20-17	16	96	0.8	65
NH-CS_25-17	16	97	1.0	64
H-CS_25-17	15	95	0.5	53
H-CS_25-34	30	91	0.7	57
H-CS_40-17	15	95	1.6	99

^a Letters in the sample names correspond to the chitosan sample used (non-hydrolyzed chitosan, NH-CS, or hydrolyzed chitosan, H-CS). The first number in sample names represents the chitosan fraction in the total solid amount, and the second number represents the total solid content (SC), in percentages.

^b Determined gravimetrically.

^c Determined gravimetrically as the percentage of grafted PVAc to the total amount of polymerized VAc, after the PVAc homopolymer was dissolved in acetone.

two reasons, (1) with a lower conversion, the ratio between CS and VAc is not constant, and (2) unreacted VAc monomer is possibly carcinogenic.

The viscosity of the adhesives prepared with NH-CS increased with increasing amount of chitosan; this was expected since chitosan gives high viscosity already at low concentrations. By using H-CS, the viscosity of the chitosan solutions, and consequently the viscosity of the adhesives, could be lowered. This can be seen when comparing adhesives with the same solid content and chitosan fraction (NH-CS_25-17 and H-CS_25-17); the viscosity of the adhesive containing H-CS was significantly lower. By increasing the solid content to 34 wt%, the viscosity was slightly increased. The highest viscosity was achieved when the chitosan fraction was 40 wt%, and the solid content 17 wt%; this showed that the amount of chitosan had a higher effect on the viscosity than the total solid content.

The grafting efficiency depends on the type of chitosan and the chitosan-VAc ratio. In the adhesives with NH-CS, the VAc grafting efficiency was the lowest in the adhesive containing the lowest chitosan fraction (NH-CS_15-17). When less chitosan (and more VAc) is present in the reaction mixture, there is a higher probability that a lower percentage of VAc will be grafted. However, NH-CS_20-17 and NH-CS_25-17 showed similar grafting efficiency, even though the ratios between

chitosan and VAc were different. Adhesives prepared with 25 wt% H-CS (H-CS_25-17 and H-CS_25-34) had lower grafting efficiencies than the adhesive containing 25 wt% NH-CS. This was hypothesized to be due to the differences in the chitosan solubility and, therefore, the stability of the system. When the chitosan fraction was further increased to 40 wt% (H-CS_40-17), the grafting efficiency was significantly increased. The chitosan:VAc ratio was changed, i.e., more chitosan was present in the adhesive; therefore, more grafting sites were likely activated along chitosan chains and a lower amount of VAc was present and available to be polymerized and grafted. Further increase of the chitosan fraction (above 40 wt%) in adhesives was not possible; preliminary results showed that due to a very high viscosity, the reaction mixture was not stirring during the polymerization reaction.

To confirm that PVAc was grafted from chitosan, CS-g-PVAc films and their acetone-soluble and acetone-insoluble fractions were analyzed with FTIR. Spectra of representative CS-g-PVAc (NH-CS_25-17 and H-CS_25-17), chitosan and the reference PVAc adhesive are shown in Fig. S2 in the Supporting information. Acetone-soluble fractions showed peaks corresponding to PVAc, and acetone-insoluble fractions showed peaks corresponding to both chitosan and PVAc.

3.3. Thermal properties

The thermal properties of chitosan (NH-CS and H-CS), prepared chitosan-graft-PVAc adhesives (CS-g-PVAc) and the reference PVAc adhesive are shown in Figs. S3–S4 in the Supporting information.

The TGA thermograms of chitosan (Fig. S3a) show that NH-CS and H-CS have similar degradation patterns, including the weight loss due to water evaporation (≈ 100 °C) and the degradation of the chitosan polymer (270 – 400 °C). This degradation trend has been reported previously [50]. A more comprehensive discussion regarding the NH-CS and H-CS thermograms can be found in the Supporting information. The PVAc thermogram (Fig. S3a) shows a two-step degradation process; (1) 320 – 370 °C, corresponding to deacetylation of the side chain, and (2) 420 – 500 °C, corresponding to the degradation of the polymer backbone. A similar degradation pattern has been reported previously [51]. TGA thermograms and 1st derivatives of CS-g-PVAc adhesives are shown in Fig. S4 in the Supporting information. The degradation pattern of CS-g-PVAc contains degradation steps characteristic for both chitosan and PVAc. Moreover, CS-g-PVAc show an additional degradation step in the range 200 – 270 °C which has been observed previously, and it has been attributed to the oxidative degradation induced by cerium ions since they can oxidize PVAc and chitosan [30]. TGA results showed that all CS-g-PVAc were stable at pressing temperature (120 °C).

DSC thermograms of chitosan (NH-CS and H-CS), prepared adhesives (CS-g-PVAc) and the reference PVAc adhesive are shown in Fig. 2. Thermograms of the CS-g-PVAc fractions soluble in acetone are shown in Fig. 2a, and the fractions insoluble in acetone are shown in Fig. 2b. DSC thermograms of adhesive films are shown in Fig. S5 in the Supporting information. CS-g-PVAc had two expected T_g s (1) between 10 and 40 °C

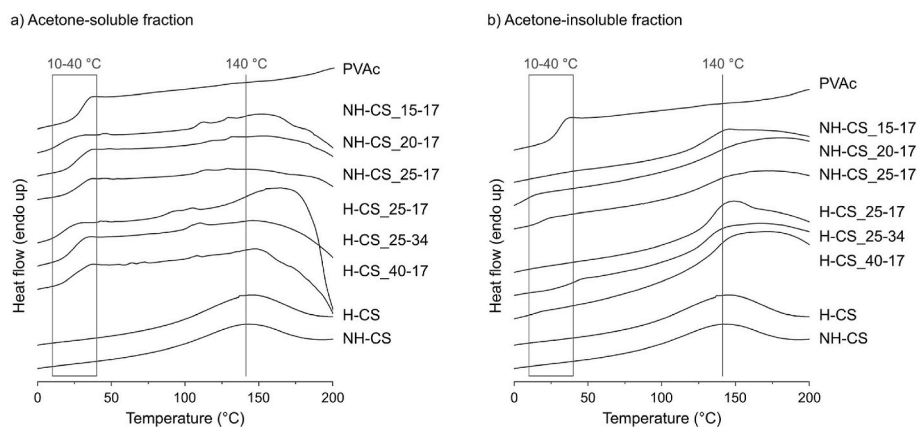


Fig. 2. DSC thermograms of non-hydrolyzed chitosan (NH-CS) and hydrolyzed chitosan (H-CS) chitosan, poly(vinyl acetate) reference (PVAc), and grafted polymers CS-g-PVAc; (a) acetone-soluble and (b) acetone-insoluble fractions of CS-g-PVAc adhesives. The first number in CS-g-PVAc sample names represents the chitosan fraction in the total solid amount, and the second number represents total solid content (SC), in percentages.

corresponding to the PVAc homopolymer, and (2) at around 140 °C corresponding to the chitosan chains. Furthermore, thermograms of the acetone-soluble fractions show several T_g s, indicating that the adhesives also contained copolymers, i.e., CS-g-PVAc (which corroborated results from the FTIR analysis).

3.4. Bond strength

The adhesive amounts applied between wood veneers, and the methods for their application (number of layers) are shown in Table S2 in the Supporting information. The applied amounts were based on the target solid spread rate (g/m^2) and adhesives solid contents. When the target solid spread rate was $61 \text{ g}/\text{m}^2$, all adhesives were applied in one layer. When the target solid spread rate was $122 \text{ g}/\text{m}^2$, adhesives with low solid contents ($\leq 17 \text{ wt}\%$) were applied in two layers with a 20 min interval to avoid over-flow. According to European Standard EN 204 [45], the dry tensile shear strength should be $\geq 10 \text{ MPa}$. In this study, the dry strength of all adhesives was lower, even the strength of the reference, commercial, PVAc adhesive. This was due to the thickness of wood substrates which was lower than the standard thickness [44]. However, even by using thin wood substrates (1.5 mm thickness), it was possible to analyze and compare different adhesive formulations. The wet tensile shear strength of specimens conditioned according to Table 2 should be $\geq 2 \text{ MPa}$.

3.4.1. Adhesives containing non-hydrolyzed chitosan (NH-CS)

Fig. 3 shows the tensile shear strengths of samples bonded with

adhesives containing non-hydrolyzed chitosan (NH-CS), with solid spread rates of 61 and $122 \text{ g}/\text{m}^2$. When comparing the dry strength of veneers bonded with CS-g-PVAc applied with the rate of $61 \text{ g}/\text{m}^2$, the dry strength (2.8–3.0 MPa) was lower than the strength of the reference adhesives, NH-CS and PVAc (5.6 and 5.2 MPa, respectively). The applied solid amount of $61 \text{ g}/\text{m}^2$ is lower than what is usually used in the wood industry, e.g., the spread rate of PVAc adhesives can be $90 \text{ g}/\text{m}^2$ ($180 \text{ g}/\text{m}^2$ with 50 wt% solid content). Therefore, all adhesives were also applied in two layers to achieve a solid spread rate of $122 \text{ g}/\text{m}^2$. With more adhesive in the bond line, the dry strength was increased with no significant differences between CS-g-PVAc formulations (5.3–5.4 MPa). All samples had wood failure after the testing; therefore, the strength of all samples was similar since it depended mostly on the strength of the wood substrates. The wet strength of the specimens bonded with CS-g-PVAc adhesives (0.4–0.8 MPa) was not significantly improved compared to the specimen bonded with a reference PVAc, when the solid spread rate was $61 \text{ g}/\text{m}^2$. However, with more adhesives in the bond lines, the water resistance was improved (1.3 MPa).

3.4.2. CS-g-PVAc with non-hydrolyzed (NH-CS) and hydrolyzed (H-CS) chitosan with the same chitosan fraction (25 wt%)

The tensile shear strength of samples bonded with adhesives containing 25 wt% non-hydrolyzed (NH-CS) and hydrolyzed chitosan (H-CS) is shown in Fig. 4. When comparing adhesives with 17 wt% solid content (NH-CS_25-17 and H-CS_25-17), the dry strength of adhesive containing NH-CS was lower than the strength of the adhesive containing H-CS (3.0 and 3.6 MPa, respectively), but this difference was not

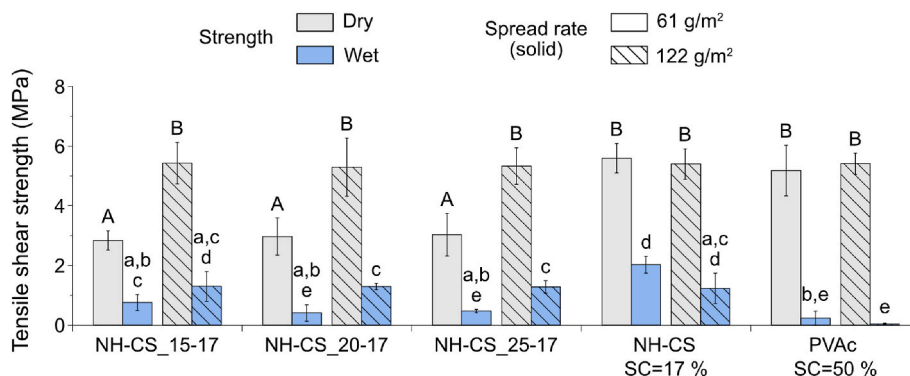


Fig. 3. Tensile shear strength of veneers bonded with CS-graft-PVAc adhesives with non-hydrolyzed chitosan (NH-CS), and reference adhesives NH-CS and poly(vinyl acetate) (PVAc). The first number in sample names represents the chitosan fraction in the total solid amount, and the second number represents the total solid content (SC), in percentages. Error bars represent standard deviations, and significant differences between samples are indicated with the different letters (uppercase for the dry strength, and lowercase for the wet strength).

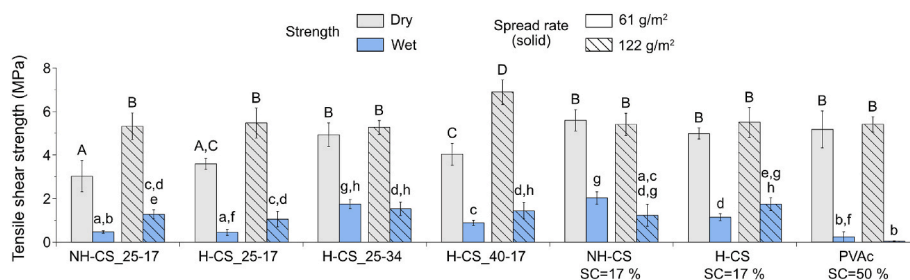


Fig. 4. Tensile shear strength of veneers bonded with CS-graft-PVAc adhesives with non-hydrolyzed chitosan (NH-CS) and hydrolyzed chitosan (H-CS), and reference adhesives NH-CS, H-CS and poly(vinyl acetate) (PVAc). The first number in sample names represents the chitosan fraction in the total solid amount, and the second number represents the total solid content (SC), in percentages. Error bars represent standard deviations, and significant differences between samples are indicated with the different letters (uppercase for the dry strength, and lowercase for the wet strength).

statistically significant. Regarding the application, the viscosity of the adhesive containing H-CS was lower, and therefore, easier to apply on wood. Interestingly, the adhesive with the same chitosan fraction, but 34 wt% solid content had the highest strength. Moreover, the wet strength of the adhesives with 17 wt% solid content was similar (0.5 MPa), but the strength of the adhesive with 34 wt% solid content was higher (1.8 MPa). The solid content, i.e., amount of chitosan in the system, could have affected the copolymer structure, which would have further affected the bond strength. In this study, the structure of the grafted polymers has not been evaluated, besides the grafting efficiency; a future study with a focus on how changes in the CS-g-PVAc structure affect the bonding properties might give more information regarding the differences in bond strengths.

With a higher solid spread rate (122 g/m²), the dry strength of all samples was similar (5.3–5.5 MPa), mostly due to the strength of the wood substrates. The wet strengths were in the range of 1.0–1.5 MPa, but without statistically significant differences.

3.4.3. Adhesive containing a higher chitosan fraction (40 wt% H-CS)

The tensile shear strength of samples bonded with the adhesive containing 40 wt% hydrolyzed chitosan (H-CS) can be seen in Fig. 4. When the chitosan fraction was increased, the solid content of the adhesive was 17 wt% (H-CS_40-17); a higher solid content was not possible to achieve due to a very high viscosity. With 61 g/m² solid spread rate, the dry strength (4.0 MPa) was similar to the strength of the adhesive containing less chitosan and the same solid content, H-CS_25-17 (3.6 MPa). However, the strength was significantly increased when the adhesive was applied with 122 g/m² (6.9 MPa). Since almost all VAc was grafted from chitosan, the molecular mobility of the chains was decreased, leading to a lower susceptibility to plastic creep in the adhesive, and possibly contributing to the final bond strength by improving stress transfer between the adhesive and the wood substrate. The strength was even higher than the strength of the PVAc reference adhesive (5.4 MPa). Future studies are needed to investigate the structure of H-CS adhesives, and especially the structure of the H-CS_40-17 adhesive, in order to understand if and how it could positively affect the wood interphase region.

4. Conclusions

In this study, poly(vinyl acetate) was covalently attached to chitosan via a grafting-from polymerization (CS-g-PVAc) in an effort to improve the water resistance and reduce the fossil-based content in wood-adhesive applications. Two chitosan samples with different molecular weights were used for the free radical grafting polymerization with VAc and the corresponding adhesives were evaluated by tensile shear strength of bonded veneers. With higher molecular weight chitosan (non-hydrolyzed), the fraction of chitosan in the adhesives was 15–25 wt % and the maximum solid content 17 wt%; a larger fraction of chitosan and a higher solid content was not possible to achieve due to very high

viscosity. The VAc grafting efficiency (percentage of grafted PVAc of total PVAc amount) was between 49 and 65 %. By using chitosan with a lower molecular weight (hydrolyzed), the solid content of the adhesive with 25 wt% chitosan could be increased to 34 wt% resulting in less water to be evaporated during pressing. Further, it was possible to increase the fraction of hydrolyzed chitosan to 40 wt% (17 wt% solid content); during grafting polymerization which also improved the grafting efficiency to 99 %.

With the non-hydrolyzed chitosan and a solid spread rate of 122 g/m², an increase in the fraction of chitosan from 15 to 25 wt% in the grafting polymerization did not significantly affect the dry bond strength (5.3–5.4 MPa) of the bonded samples since it was mostly limited by the strength of the wood substrate. However, with 25 wt% chitosan, the viscosity was inappropriately high. For the hydrolyzed chitosan samples the viscosity was lower but adhesives with 25 wt% hydrolyzed chitosan and a spread rate of 122 g/m² had similar dry strength (5.3–5.5 MPa) as the non-hydrolyzed samples. Interestingly, the adhesive with 40 wt% hydrolyzed chitosan (H-CS_40-17) had a higher dry strength (6.9 MPa), even higher than the reference PVAc adhesive (5.4 MPa) despite very high viscosity. It is noteworthy that the VAc grafting efficiency in the 40 wt% chitosan sample was almost 100 %, suggested to result in a more efficient stress transfer across the wood-adhesive interface and/or a strengthened wood interphase which has to be further investigated in future studies. I.e., this sample, H-CS_40-17, had the highest grafting efficiency, the highest strength and the highest biobased content in this study.

This study demonstrates that copolymers of chitosan and VAc can be used as more environment-friendly and durable wood adhesives, with significantly decreased fossil content, and improved water stability, compared with commercial PVAc adhesives. Optimization of the chitosan molecular weight and polymerization parameters could further improve the overall performance and allow for further reduction of the fossil-based content. Additional studies are needed to investigate the effect of the structure of CS-g-PVAc adhesives on the bond performance and an assessment on what environmental-related benefits the partly biobased adhesive may bring.

CRedit authorship contribution statement

Tijana Todorovic: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation. **Emelie Norström:** Validation, Methodology, Investigation. **Linda Fogelström:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Eva Malmström:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijadhadh.2024.103818>.

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